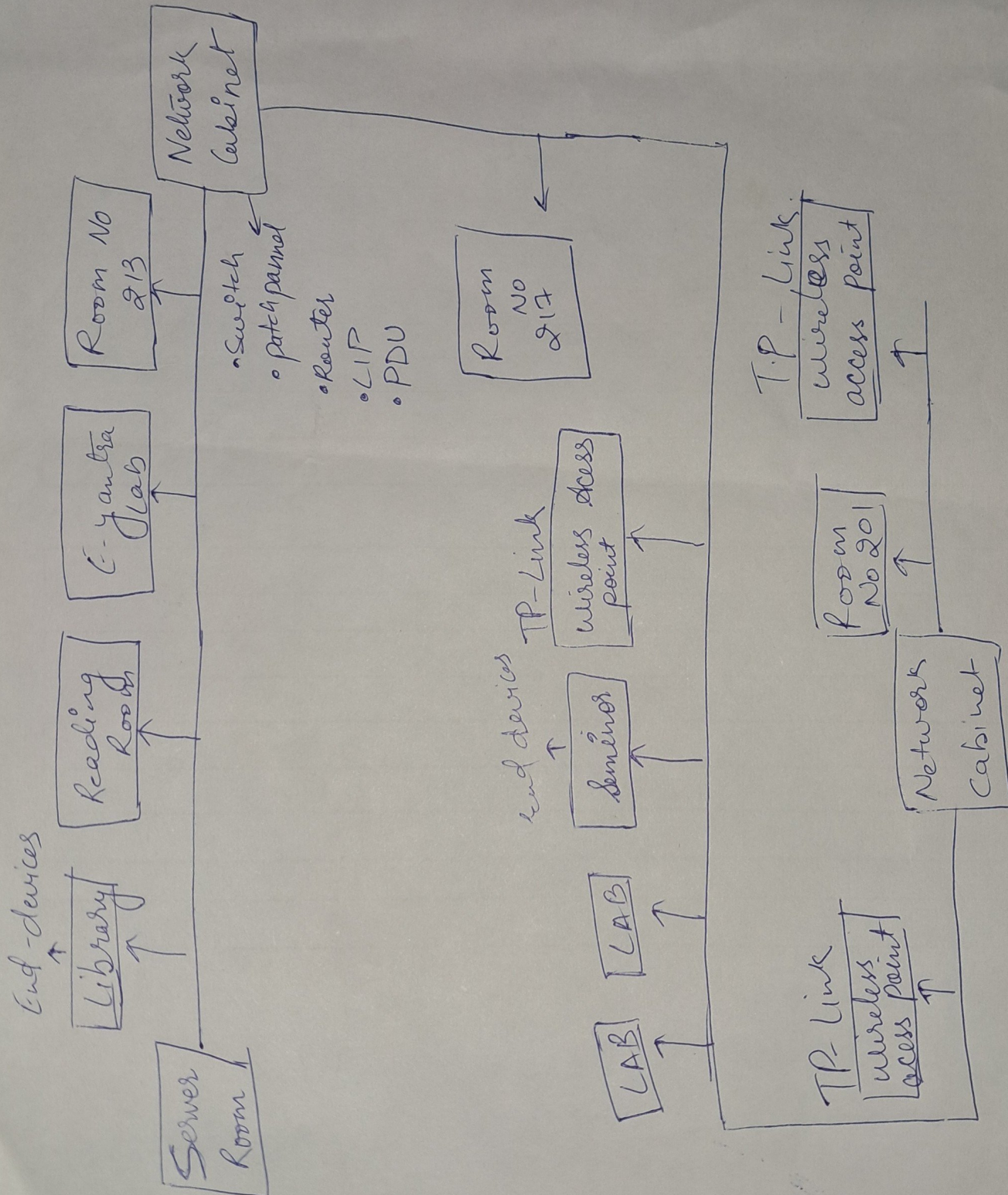


1. Identify and Draw LAN Topology of your organization
- Clearly label all components such as Router, Switch, Access point and end devices.



② Identify Network Devices.

- List all network devices used in LAN
- Mention the function of each devices

→

(i) Network Switch:-

- It connects multiple devices (computers, printers, access points) within same local network
- It primarily operates in layer 2 (Data-link layer) of the OSI Model and uses MAC address to forward data directly to the specific destination devices

(ii) Router:-

- Connects different network together (such as connecting your LAN to the internet or connecting multiple subnetworks).

(iii) Wireless access point (WAP):-

- Connects wireless devices (Smartphones, Laptop, tablets) to a wired network using Wi-Fi radio signals.

(iv) Network interface card (NIC)

- The hardware component installed in a device (or integrated into its motherboard) that allow it to physically connect to a network via an Ethernet cable or wireless via wifi

(v) Patch pannel

- A passive networking component used to organize, manage and terminate incoming and outgoing Ethernet cable in a central location.

(vi) Modem (Modulator - Demodulator)

→ Converts signals from an internet service provider (ISP) - Such as fiber, coaxial cable or DSL - into digital data that local network devices (like a router) can understand

(vii) Firewall

- Monitors and filters incoming and outgoing network traffic based on predefined security rules
- It acts as a security barrier b/w the trusted internal LAN and untrusted external network

(viii) Network Hub

→ An older simpler device that connects multiple computers in a network unlike a switch, it broadcasts all incoming data packets to every connected port which can cause network congestion and security.

③ Analyze Connection Type:-

a) Specific wheather connections are:-

- Wired (Ethernet) - for classroom smartboard, library computers.
- Wireless - for student smartphone & laptop.

Wireless LAN adapter Wi-Fi: `ipconfig /all`

```
Connection-specific DNS Suffix . :  
Description . . . . . : MediaTek Wi-Fi 6 MT7921 Wireless LAN Card  
Physical Address. . . . . : CC-47-40-C7-28-D3  
DHCP Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . . : fe80::688a:8c66:92c3:b629%4(Preferred)  
IPv4 Address. . . . . : 10.85.20.80(Preferred)  
Subnet Mask . . . . . : 255.255.248.0  
Lease Obtained. . . . . : 26 July 2026 16:17:51  
Lease Expires . . . . . : 27 July 2026 04:21:38  
Default Gateway . . . . . : 10.85.16.1  
DHCP Server . . . . . : 10.85.16.1  
DHCPv6 IAID . . . . . : 214714176  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-40-07-73-08-BF-B8-DA-AB-DB  
DNS Servers . . . . . : 8.8.8.8  
                        1.1.1.1  
NetBIOS over Tcpip. . . . . : Enabled
```

```
PS C:\WINDOWS\system32> Get-NetAdapter
```

Name	InterfaceDescription	ifIndex	Status	MacAddress	LinkSpeed
----	-----	-----	-----	-----	-----
Ethernet	Realtek PCIe GbE Family Controller	10	Disconnected	08-BF-B8-DA-AB-DB	0 bps
Bluetooth Network Conn...	Bluetooth Device (Personal Area Netw...	9	Disconnected	CC-47-40-C7-28-D2	3 Mbps
Wi-Fi	MediaTek Wi-Fi 6 MT7921 Wireless LAN...	4	Up	CC-47-40-C7-28-D3	78 Mbps

⑥ Briefly Explain.

a) Difference b/w wired and wireless speed.

feature	Wired speed. (Ethernet)	Wireless speed (Wi-fi)
Max real world speed	Consistently High	Variable
Latency (Ping)	Ultra-low	Higher and fluctuating
Stability / Packet loss	Near 0% packet loss	Subject to packet loss from physical obstacles and frequency congestion
Band width. Distribution	full dedicated bandwidth per physical cable connection	shared total band width among all wireless connected devices
Distance impact	Constant speed up to 100 mb	Speed drops significantly as a distance from AP/ Router increases.

⑥ Role of Subnet mask in networking.

- Differentiating Network ID from Host ID
- Determine Local vs Remote Traffic
- Reduce Broadcast Congestion.
- Enables Network Segmentation and Security
- Optimizes IP Address Allocation.